KAIZEN MATHS

10-MINUTE PRACTICE

Algebraic substitution

Single variable - mixed practice

Mix linear and quadratic substitution in one short practice session.

10 MINUTES / 8 QUESTIONS / AS-09

Name _____ Date _____

Show your working. Use brackets for negative values.

1 Given $x = 6$

Evaluate $7x + 10$

2 Given $x = -3$

Evaluate $x^2 + 4x - 8$

3 Given $x = 6$

Evaluate $7x + 2$

4 Given $x = 0$

Evaluate $x^2 + 3x + 4$

5 Given $x = 1$

Evaluate $6x - 8$

6 Given $x = 5$

Evaluate $x^2 + 3x - 4$

7 Given $x = 8$

Evaluate $6x - 8$

8 Given $x = -3$

Evaluate $x^2 + 6x + 1$

Answer sheet

Single variable - mixed practice | 8 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 52

02 -11

03 44

04 4

05 -2

06 36

07 40

08 -8

Worked solutions

Single variable - mixed practice

01 Given $x = 6$

$7x + 10$

Substitute $x = 6$.

$7\times 6 + 10$

$42 + 10$

52

02 Given $x = -3$

$x^2 + 4x - 8$

Substitute $x = -3$.

$(-3)^2 + 4\times(-3) - 8$

$9 - 12 - 8$

-11

03 Given $x = 6$

$7x + 2$

Substitute $x = 6$.

$7\times 6 + 2$

$42 + 2$

44

04 Given $x = 0$

$x^2 + 3x + 4$

Substitute $x = 0$.

$(0)^2 + 3\times 0 + 4$

$0 + 4$

4

Worked solutions

Single variable - mixed practice

05 Given $x = 1$

$6x - 8$

Substitute $x = 1$.

$6\times 1 - 8$

$6 - 8$

-2

06 Given $x = 5$

$x^2 + 3x - 4$

Substitute $x = 5$.

$(5)^2 + 3\times 5 - 4$

$25 + 15 - 4$

36

07 Given $x = 8$

$6x - 8$

Substitute $x = 8$.

$6\times 8 - 8$

$48 - 8$

40

08 Given $x = -3$

$x^2 + 6x + 1$

Substitute $x = -3$.

$(-3)^2 + 6\times(-3) + 1$

$9 - 18 + 1$

-8